Read, Not Reasoned? Ground-Truth Causal Interventions on Multi-Hop Reasoning Chains

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Abstract

When a language model writes a reasoning chain, does its answer depend on that chain, and through which path? We study this with interventions whose ground truth is known by construction: inserted redundant hops (true necessity 0), programmatic input changes (true effect 0), and an exact mediation analysis in which the chain is set directly. We report four results on two multi-hop datasets with four models from two families (Qwen2.5 0.5B–3B, SmolLM2-1.7B). **(i) Step masking is blind to bridge reasoning.** Masking one intermediate hop changes p(gold) by at most 4 points, whereas masking it together with its descendants in the chain's reasoning graph changes it by 49–77 points; we show this is an identity, not noise (single-step masking estimates a controlled direct effect, which vanishes on bridge chains). **(ii) Resampling-based necessity with a fixed gold baseline is biased by the generator.** Probability-of-necessity estimates computed against the original chain assign 55–80 points of necessity to provably redundant steps; the bias equals one minus the rate at which the model can regenerate a correct continuation, and disappears with a resampled baseline. **(iii) Up to 3B, chains are read, not reasoned over.** Answers depend on the chain when the answer is a string written in it (56–77 points) and barely at all when it must be composed from it ($\leq$ 2 points up to 1.7B, 7 at 3B, the level of the placebo); appending one hop that states the answer restores the dependence at every scale. **(iv) Distractors harm chain-of-thought through the chain.** With the chain held at its original value the effect of four distractor paragraphs vanishes (direct effect +1); the whole loss (−9 EM) is mediated by the regenerated chain. A readout-level decoding probe that contrasts the model with interventions on its own hops therefore cannot help, and does not. All hypotheses were fixed before each GPU run.

1 Introduction

Chain-of-thought prompting [28] makes models write intermediate steps, but writing a step is not the same as *using* it. Causal tests show that models frequently ignore or post-rationalise their chains [6, 12, 24, 25], and that step importance depends on how a step is removed [2, 9]. What these studies lack is ground truth: without knowing which steps are necessary, an estimator of step importance cannot be validated, only compared with other estimators.

We build that ground truth. On CLUTRR [18] (kinship graphs) and MuSiQue [23] (real multi-hop questions with gold decompositions) we write gold reasoning chains, insert one true but irrelevant hop, and ask each estimator to separate necessary from redundant hops. We pair this with binary input interventions whose effect on the answer is zero by construction, placebo and mask-token controls, and an exact mediation analysis in which the chain itself is set, so that direct and indirect effects are observed per question rather than estimated.

Contributions.

1. **A ground-truth protocol for chain reliance**: necessity labels and null effects known by construction; placebo, mask-marker and volume-matched controls; exact NDE/TIE mediation.
2. **Two estimator biases, with closed forms.** Single-step masking measures a controlled direct effect that vanishes on bridge chains (proposition 1); resampling necessity against a fixed gold continuation adds $1 - r$, where r is the model's regeneration success (proposition 2). Both are confirmed at every scale we test.
3. **Read, not reasoned (up to 3B).** Reliance on the chain appears when the answer can be copied from it and is near zero when it must be composed; a single binary intervention (T_{copy}) switches it on.
4. **Where robustness failures live.** Distractors harm chain-of-thought entirely through the regenerated chain, which explains why readout-level contrastive decoding—including a graph-aware rule we formulate as a probe (HGCD)—cannot help at this scale.

2 Related Work

Faithfulness of reasoning chains. Lanham et al. [6] intervene on model-written chains (truncation, inserted mistakes, paraphrases); Turpin et al. [24] show biased features steer answers without being verbalised; FRODO [12] uses causal mediation [26] and trains small models with counterfactual preferences; FUR [25] measures faithfulness by unlearning steps; Thought Anchors [2] measures step importance by counterfactual resampling, and Thought Branches [9] argues that step effects must be measured by resampling the continuation (a total effect), noting that removing a sentence while preventing its content from reappearing changes conclusions. FRODO already treats the rationale as a mediator and finds weak indirect effects for models under 20B, and Somov et al. [19] find that models often fail to update after edits to intermediate structures. Our contribution relative to these is narrower and specific: (i) necessity labels and null effects that are *known by construction*, so step-importance estimators can be *validated* rather than compared with each other; (ii) a deterministic, rollout-free total-effect intervention for hop-structured chains, validated against resampling; (iii) the bridge-versus-compositional distinction, which predicts exactly where direct-effect estimators fail; and (iv) an exact mediation of distractor effects (chain vs. readout).

Necessity and sufficiency. Yu et al. [30] define step-level probabilities of necessity and sufficiency for chain-of-thought, prove identifiability under exogeneity and monotonicity (building on Pearl [14], Tian and Pearl [22]), and use PNS to prune chains for in-context learning and fine-tuning. We show the same quantity arises as the limit of a decoding rule, and we separate direct from total effects [13].

Contrastive and guided decoding. Contrastive decoding [8, 11], DoLa [3], CFG [16], CAD and its adaptive/gated variants [17, 21, 27] contrast distributions under different models, layers or contexts. Visual contrastive decoding [7] uses a *distorted* rather than empty input; our L1 variant is the textual analogue with a type-matched counterfactual. Pced [4] treats each retrieved document as an expert contrasted with a no-context baseline; RED [29] multiplies image- and rationale-conditional distributions in vision-language models. None of these intervene on the hops of the model's own reasoning graph, distinguish direct from total effects, or relate the contrast to PNS.

Table 1: Positioning against closely related work. "GT" = the true value of the measured effect is known by construction.

Work	What is intervened on	Effect measured	GT?	Training	Controls
Lanham et al. [6]	model's CoT (truncate, add mistake, paraphrase)	answer change	no	none	paraphrase, filler tokens
Stolfo et al. [20] (ACL)	input factors of math problems	total / direct causal effect	partly	none	–
Paul et al. [12] (EMNLP-F)	rationale	mediation (indirect effect)	no	preference training	–
Yu et al. [30] (NeurIPS)	CoT step, replaced + rolled out	PNS	no (LLM judge)	SFT / ICL	–
Macar et al. [9] (ICLR)	sentence, resampled continuation	total effect, resilience	no	none	on- vs. off-policy
Somov et al. [19]	intermediate structure (rubric)	decision update rate	yes (deterministic map)	none	–
Tutek et al. [25] (EMNLP)	step information in weights	prediction change	no	unlearning	–
Shi et al. [17], Wang et al. [27]	whole context	(implicit) total context effect	no	none (decoding)	–
Ours	hop + DRG descendants; binary input interventions	direct vs. total effect; NDE/NIE	**yes**	none	placebo, mask-marker, redundant hop, content-free

Causal analyses of reasoning. Stolfo et al. [20] quantify robustness of mathematical reasoning through interventions on a causal graph of the input; Zečević et al. [31] argue models "talk" causality without being causal; text-intervention effect estimation [5] uses LLM rewrites as treatments. Our interventions are programmatic, so their ground-truth effects are known exactly.

3 Setup: a Structural Causal Model of Multi-Hop Reasoning

Variables. A question Q, a context $P = (P_{\text{sup}}, P_{\text{dist}})$ of supporting and distractor passages, a chain of hops $H_1, \ldots, H_k$, and an answer A. Each hop is rendered as `Hop` i`:` ⟨`body`⟩ `=>` ⟨`result`⟩. The model acts as a *reader* mechanism $\pi(a \mid Q, P, h_{1:k})$ over a finite candidate set $\mathcal{A}$.

Two task graphs. In *bridge* chains (MuSiQue), hop j resolves an entity that the next hop consumes and the answer is the result of the last hop: $Q \to H_1 \to \cdots \to H_k \to A$, with $P_{\text{sup},j} \to H_j$. In *compositional* chains (CLUTRR), every hop contributes a relation that is composed into the answer: $H_j \to A$ for all j and $H_j \to H_{j+1}$ through the shared entity. Distractors have no edge to A in either task graph (fig. 1).

Directed reasoning graph (DRG). Given a generated chain $y = (h_1, \ldots, h_k)$ we add an edge $i \to j$ $(i < j)$ if the result of h_i is mentioned in the body of h_j. Terminal hops are those whose result is the query target (CLUTRR) or the last hop (MuSiQue); the *relevant set* $\mathcal{R}(y)$ contains the terminal hops and their DRG ancestors. Hops outside $\mathcal{R}(y)$ have no directed path to the answer.

Interventions on hops.

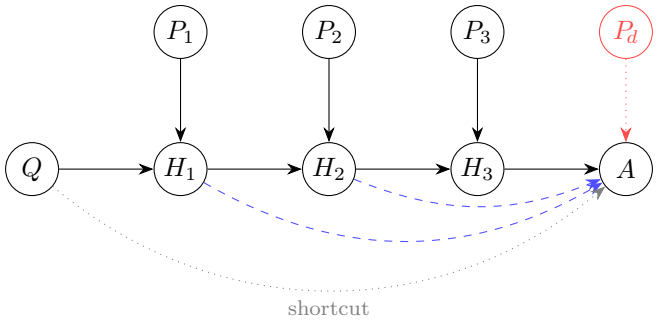


Figure 1: Task SCM. Solid: bridge chain. Dashed blue: extra edges of compositional chains. Red dotted: spurious distractor path that the *reader* may use but the task graph forbids. Gray: parametric shortcut. CAD's contrast includes the red path; HGCD holds P fixed and contrasts interventions on H_j only.

Definition 1 (Naive and graph-aware interventions)**.** *y_j^{nai} replaces hop j by a mask token and leaves all other hops unchanged. y_j^{tot} masks hop j and, in every DRG descendant d of j, masks every mention of the results of j and of d. The latter is a deterministic, rollout-free approximation of* $\text{do}(H_j{=}\perp)$*: descendant values are functions of h_j, hence unknown under the intervention. A* placebo *rewrites hop j without changing its entities or relation.*

For an answer a we write $\Delta_j^{\text{nai}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{nai}})$ and $\Delta_j^{\text{tot}}(a) = \pi(a \mid y) - \pi(a \mid y_j^{\text{tot}})$, suppressing Q, P.

What is (and is not) causal here. We do not assume the language model *is* an SCM of the task. The reader π is a deterministic function of its input, so editing hops in the prompt is an exact intervention on the reader's input, and Δ_j is an exact (not estimated) effect *on the reader*. The task SCM only supplies ground truth: which hops are necessary, and which input changes should have no effect on the answer.

Mask-token artifacts. Replacing text by `[MASKED]` is itself out of distribution. Our estimator benchmark controls for this: the inserted redundant hop is masked in exactly the same way, so any effect caused by the mask token alone appears as a non-zero Δ on redundant hops and lowers the AUC. In addition, a *mask-marker* control inserts `[MASKED]` into hop j while keeping all of its content; its effect Δ_j^{marker} measures the mask-token artifact directly and should be ≈ 0. Because y_j^{tot} masks a whole subtree, we also use a *volume-matched* control that inserts the same mask tokens into hop j and every descendant while keeping all of their content, and a counterfactual-replacement version of the total effect that uses no mask on hop j at all.

4 Theory: two estimator biases

Assumption 1 (Markov reader on bridge chains)**.** *For bridge chains, $\pi(a \mid h_{1:k})$ depends on $h_{1:j}$ only through $h_{j+1:k}$ whenever h_{j+1} restates the result of h_j $(j < k)$.*

Proposition 1 (Single-step masking is a direct effect)**.** *$\Delta_j^{\text{nai}}(a)$ equals the controlled direct effect of H_j on A with $H_{j+1:k}$ held at their observed values. Under assumption 1, $\Delta_j^{\text{nai}}(a) = 0$ for every $j < k$ on bridge chains, whereas $\Delta_j^{\text{tot}}(a)$ is not constrained to zero.*

Proposition 2 (Fixed-baseline resampling is biased by the generator)**.** *Let q_j be the probability of a correct answer when hop j is corrupted and the continuation is resampled, and r_j the same probability when hop j is kept and the continuation is resampled. On chains answered correctly with the original continuation (*$\mathrm{PS} = 1$*), the fixed-baseline estimator $1 - q_j$ equals $(r_j - q_j) + (1 - r_j)$: the resampled-baseline effect plus a bias $1 - r_j \geq 0$ that does not depend on whether hop j is necessary. For a redundant hop ($r_j = q_j$) it returns $1 - r_j$ instead of 0.*

Remark. The identifiability lemma of Yu et al. [30] assumes $P(A{=}y \mid \mathrm{do}(S)) = 1$, i.e. $r_j = 1$; proposition 2 quantifies what happens when the rollout model cannot regenerate a correct continuation. Thought-anchor style resampling [2] uses a resampled baseline and is unbiased in this sense.

5 Experimental Protocol

Data. CLUTRR (`gen_train234_test2to10`), restricted to clean simple-path stories (11,708 train / 635 test / validation), gives exact gold paths. **MuSiQue-Ans** gives real text with gold decompositions; the context holds the supporting paragraphs plus 4 distractors. MuSiQue was built to resist single-hop shortcuts [10].

Estimator benchmark. For each gold chain we insert one true but off-path hop (necessity 0) and score every hop with: mask hop only (Δ^{nai}); mask hop plus DRG descendants (Δ^{tot}); counterfactual replacement of the hop's result without and with propagation; resampling necessity with a fixed gold baseline and with a resampled baseline ($K = 2$ samples per arm); and three controls (placebo rewrite, mask marker on the hop, mask markers on the whole subtree). Δ is measured on the gold answer's probability normalised over the candidate set and, as a robustness check that needs no candidate set, on its raw log-probability. Per model we use 180 CLUTRR and 150 MuSiQue gold chains (over 2,000 scored hops including the T_{copy} variant).

Copy vs. compute. $T_{\mathrm{copy}} \in \{0, 1\}$ appends to a CLUTRR gold chain one hop that states the answer relation; everything else is identical. A second variant states a *wrong* relation that contradicts the relational hops; we report how often the model follows it, separately for chains the model solves without the extra hop.

Input interventions and mediation. `rename`, `order`, `distract` (CLUTRR) and `dist8`, `order` (MuSiQue) have true effect 0. For each question we observe $Y(0, M(0))$ (original input, original chain), $Y(1, M(1))$ (intervened input, regenerated chain) and $Y(1, M(0))$ (intervened input, *original* chain fed back), giving the natural direct effect $\mathrm{NDE} = \mathbb{E}[Y(1, M(0)) - Y(0, M(0))]$ and the total indirect effect $\mathrm{TIE} = \mathrm{TE} - \mathrm{NDE}$ [13] without cross-world assumptions.

Models and statistics. Qwen2.5-0.5B/1.5B/3B-Instruct [15] and SmolLM2-1.7B-Instruct [1], fp16, greedy decoding, one Kaggle T4 each. Bootstrap 95% CIs (2,000 resamples), paired bootstrap and exact McNemar tests.

6 Results

All tables are produced by the released code; nothing is typed by hand.

Table 2: Effects on p(gold) by hop position across scale (points; estimator benchmark, Qwen2.5 and SmolLM2). Parsed from the run log by `tools/parse_reliance_log.py`.

Quantity (Δp(gold), points)	qwen-0.5b	qwen-1.5b	smollm2-1.7b	qwen-3b
CLUTRR, non-final gold hop: total	0.1	-1.5	1.8	7.3
CLUTRR, redundant hop: total	-1.9	-0.2	-0.4	2.0
CLUTRR, final gold hop: placebo	0.7	-0.0	4.1	7.4
CLUTRR+copy, non-final: total	60.8	76.1	69.5	77.0
CLUTRR+copy, non-final: direct	1.3	1.3	1.0	1.7
MuSiQue, non-final: total	56.7	56.9	66.8	49.1
MuSiQue, non-final: direct	-2.4	0.6	-3.9	0.5
MuSiQue, final: total	67.2	69.7	73.5	55.0
Rollout PNS, gold baseline: necessary (non-final)	78.1	74.4	72.2	60.8
Rollout PNS, gold baseline: *redundant*	79.5	74.5	74.7	55.3

Table 3: Main quantities for the two models with full decoding runs.

Quantity	qwen-1.5b	smollm2-1.7b
Content effect of chain, CLUTRR	0.8	2.0
Content effect, CLUTRR + answer hop (T_{copy}=1)	75.1	62.8
Content effect, MuSiQue	56.4	71.9
CLUTRR+copy non-final hops: Δ^{tot} / Δ^{nai}	76.3 / 1.3	68.5 / 1.1
MuSiQue non-final hops: Δ^{tot} / Δ^{nai}	66.3 / 0.7	72.8 / -3.6
AUC MuSiQue: total / direct	0.975 / 0.828	0.988 / 0.682
AUC CLUTRR+copy: total / direct	1.000 / 0.658	0.991 / 0.643
MuSiQue `dist8`, CoT: TE	-9.2	-7.5
NDE (chain fixed) / NIE (via chain)	0.8 / -10.0	1.7 / -9.2
MuSiQue `dist8`, direct answer: NDE	-4.2	-6.7
CLUTRR accuracy: calibration − CoT	5.9	3.4

R1: single-step masking misses bridge reasoning (proposition 1). Wherever the answer is written in the chain (MuSiQue; CLUTRR with T_{copy}), masking one non-final hop moves p(gold) by -3.9 to $+1.7$ points at every scale, while masking it with its descendants moves it by 49–77 points, close to the final hop's own effect (table 2). Total-effect estimators reach AUC 0.975–1.000 and direct-effect ones 0.64–0.83 (table 4); the counterfactual-replacement implementation, which puts no mask on hop j, agrees with the masking one, and placebo and mask-marker controls are at chance.

R2: fixed-baseline resampling assigns necessity to redundant steps (proposition 2). On MuSiQue the fixed-baseline resampling estimator gives 55–80 points of necessity to the *redundant* hop, indistinguishable from necessary hops (61–78; table 2). By proposition 2 this is $1 - r$, i.e. the models regenerate a correct continuation only 20–45% of the time. The resampled-baseline estimator removes the term by construction.

R3: chains are read, not reasoned over, up to 3B. On CLUTRR the answer is a relation that must be composed from the hops and never appears in them. The total effect of a non-final gold hop is -1.5 to 1.8 points up to 1.7B and 7.3 at 3B, where the placebo control on the final hop is equally large (7.4), so no reliable dependence is detectable. The models also fail to compose from a perfect chain (correct on about 19% of gold chains at 1.5B), and chains that reach every

gold-path entity do not raise CoT accuracy (table 6). Appending one hop that states the answer (T_{copy}) produces the full bridge pattern at every scale (non-final total 61–77, direct 1.0–1.7). The dependence on the chain therefore tracks the readout the model can execute—copying a string—rather than the reasoning the chain encodes. Whether composition over the chain appears at 7B and above is the open question of section 7.

R4: distractors act through the chain. On MuSiQue, four extra distractor paragraphs lower CoT exact match by 9.2 (Qwen-1.5B) and 7.5 (SmolLM2) points. Feeding the original chain back under the distracted input removes the effect (NDE +0.8 and +1.7); it is entirely mediated by the regenerated chain (TIE −10.0 [−16.7, −4.2] and −9.2 [−15.8, −2.5]; table 5). Every chain-based reader shows the same split. We do *not* find that CoT is more vulnerable than direct answering overall: the paired difference of their total effects is −5.8 [−11.7, 0.0] and −1.7 [−6.7, 3.3] points.

R5: a readout-level decoding probe cannot help, as R3–R4 predict. We formulate HGCD, which adds α times the average total effect of the chain's answer-relevant hops to each candidate's probability (appendix A). It can only amplify a dependence that exists and acts after the chain is generated. Consistently, it equals CoT on both datasets and models ($|\Delta| \leq 1.7$ points, $p \geq 0.6$). The only decoding rule that beats CoT on CLUTRR is content-free label calibration [32] (+5.9 points for Qwen-1.5B), which uses no information from the chain and is itself fragile under a neutral-sentence placebo (−5.0 [−10.1, −0.4] for SmolLM2). The mixture-reader assumption behind our decoding propositions (appendix A) is not supported by these data.

Table 4: Estimator benchmark, Qwen2.5-1.5B (AUC for necessary vs. redundant hops).

Estimator	clutrr		clutrr_copy		musique	
	all	PS=1	all	PS=1	all	PS=1
Mask hop only (direct)	0.532	0.583	0.658	0.664	0.828	0.838
Mask hop + DRG descendants (total)	0.523	0.539	1.000	1.000	0.975	0.976
Replace result, keep downstream (direct)	0.582	0.650	0.672	0.670	0.764	0.767
Replace result, propagate (total)	0.576	0.635	1.000	1.000	0.978	0.978
DRG path indicator (structure only)	1.000	1.000	0.500	0.500	0.994	0.994
Placebo rewrite (control)	0.468	0.469	0.498	0.509	0.456	0.450
Mask token, content kept (control)	0.542	0.558	0.501	0.515	0.505	0.517

7 Limitations and scope

Scale. Our models are 0.5–3B. R1 and R2 are identities whose magnitude we confirm at every scale; R3 is a claim about this range only. A 7B run (Qwen2.5-7B, OLMo-2-7B) and a 14B run are the natural next step to look for the onset of composition over the chain. **Sample size.** The estimator benchmark scores about 1,300 hops per model, so position means are tight; the decoding comparisons use 238 CLUTRR and 120 MuSiQue questions, so accuracy differences below about 6 points are not resolved, and we make no claims that depend on them. **Assumptions.** assumption 1 idealises the reader; the data match its prediction. The mixture assumption used for the decoding probe does not hold (R5). **Masking.** Subtree masking inserts several mask tokens; counterfactual replacement and the volume-matched marker control address this. **Parsing.** The DRG is built from explicit mentions. On the models' own CLUTRR chains, format success is 100%, gold-path entities are reached in 57% (Qwen) / 60% (SmolLM2) of hops, and the DRG is fully connected

Table 5: Mediation on MuSiQue, Qwen2.5-1.5B (EM points; NIE column = TE − NDE = TIE).

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-3.3 [-10.0,2.5]	-4.2 [-10.0,0.9]	0.8 [-1.7,4.2]
dist8	CoT	-9.2 [-16.7,-2.5]	0.8 [-3.3,5.0]	-10.0 [-16.7,-4.2]
dist8	CAD	-0.8 [-7.5,5.8]	-1.7 [-7.5,4.2]	0.8 [-3.3,5.0]
dist8	AdaCAD	-0.8 [-8.3,5.8]	-2.5 [-9.2,3.3]	1.7 [-2.5,5.8]
dist8	CAD-chain	-5.8 [-13.3,0.8]	1.7 [0.0,4.2]	-7.5 [-15.0,-0.8]
dist8	Calibration (content-free)	-9.2 [-17.5,-1.7]	0.8 [0.0,2.5]	-10.0 [-18.3,-2.5]
dist8	Direct × CoT (PoE)	-8.3 [-15.8,-1.7]	1.7 [-3.3,6.7]	-10.0 [-17.5,-3.3]
dist8	HGCD (no graph)	-6.7 [-14.2,0.8]	2.5 [-0.8,5.8]	-9.2 [-16.7,-2.5]
dist8	HGCD (log-space)	-7.5 [-15.0,-0.8]	0.8 [0.0,2.5]	-8.3 [-15.8,-1.7]
dist8	HGCD (adaptive α)	-7.5 [-15.8,0.0]	1.7 [-1.7,5.0]	-9.2 [-16.7,-2.5]
dist8	**HGCD**	-7.5 [-15.0,0.0]	1.7 [-1.7,5.0]	-9.2 [-16.7,-2.5]
order	Direct (no CoT)	-4.2 [-10.8,2.5]	-3.3 [-9.2,2.5]	-0.8 [-4.2,2.5]
order	CoT	0.0 [-6.7,6.7]	-1.7 [-5.0,1.7]	1.7 [-5.0,8.3]
order	CAD	-4.2 [-11.7,3.3]	-1.7 [-8.3,5.0]	-2.5 [-7.5,2.5]
order	AdaCAD	-4.2 [-11.7,3.3]	-2.5 [-9.2,3.3]	-1.7 [-6.7,3.3]
order	CAD-chain	0.8 [-5.8,8.3]	0.0 [0.0,0.0]	0.8 [-5.8,8.3]
order	Calibration (content-free)	0.8 [-7.5,8.3]	0.0 [-2.5,2.5]	0.8 [-6.7,8.3]
order	Direct × CoT (PoE)	-3.3 [-10.0,4.2]	-3.3 [-9.2,3.3]	0.0 [-5.8,5.8]
order	HGCD (no graph)	4.2 [-3.3,11.7]	-0.8 [-2.5,0.0]	5.0 [-2.5,12.5]
order	HGCD (log-space)	0.0 [-7.5,6.7]	-0.8 [-2.5,0.0]	0.8 [-5.8,7.5]
order	HGCD (adaptive α)	2.5 [-5.8,10.8]	-2.5 [-5.8,0.0]	5.0 [-2.5,12.5]
order	**HGCD**	2.5 [-5.0,10.0]	-1.7 [-4.2,0.0]	4.2 [-3.3,11.7]

Table 6: CLUTRR CoT accuracy by whether the model's own chain reaches every gold-path entity.

Method	gold-path chain (n=41)	other chain (n=197)
CoT	17.1 [7.3,29.3]	18.3 [13.2,23.9]
Self-consistency	9.8 [2.4,19.5]	23.4 [17.3,29.4]
CAD-chain	19.5 [7.3,31.7]	17.8 [12.7,23.4]
HGCD (naive mask)	22.0 [9.8,34.1]	23.4 [17.3,28.9]
HGCD	17.1 [7.3,29.3]	18.3 [13.2,23.9]

in 25% / 36% of chains; this affects only the decoding probe, since the estimator benchmark uses gold chains. **What the reported runs contain.** The scale study (table 2) was run before we added the resampled-baseline estimator, the wrong-answer variant of T_{copy}, the volume-matched mask control and the raw log-probability effect; these are implemented and tested on CPU, and their GPU results are not yet included. **Candidate sets.** Effects on MuSiQue are measured on the gold answer's probability over a candidate set (gold recall 56% before adding the model's strings); the raw log-probability version of the effect needs no candidate set.

8 Conclusion

With interventions whose ground truth is known, two common ways of measuring the importance of a reasoning step turn out to be biased in predictable ways: single-step masking measures a direct effect that vanishes on bridge chains, and fixed-baseline resampling adds the generator's failure rate. Correctly measured, models up to 3B depend on their chains when the answer can be read from

them, not when it must be computed, and distractors hurt chain-of-thought through the chain it regenerates. Decoding rules that act on the readout cannot repair either.

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Algorithm 1 HGCD (L2) for one question

$y \leftarrow$ greedy chain from the model (optionally with L1 contrast on result tokens)
Parse hops; build DRG; compute relevant set $\mathcal{R}(y)$
$p \leftarrow \pi(\cdot \mid Q, P, y)$ over $\mathcal{A}$ ▷ one forward + KV-cached candidates
for $j \in \mathcal{R}(y)$ **do**
    $p_j \leftarrow \pi(\cdot \mid Q, P, y_j^{\text{tot}})$
end for
return $\arg\max_{a \in \mathcal{V}_\beta} (1+\alpha) p(a) - \alpha \operatorname{mean}_j p_j(a)$

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A The decoding probe in full

Answer-level rule (L2). With uniform weights $w_j = 1/|\mathcal{R}(y)|$ on relevant hops and a plausibility set $\mathcal{V}_\beta = \{a : \pi(a \mid y) \geq \beta \max_{a'} \pi(a' \mid y)\}$ [8],

$$\hat{a} = \arg\max_{a \in \mathcal{V}_\beta} s_\alpha(a),$$
$$s_\alpha(a) = (1+\alpha)\,\pi(a \mid y) - \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \pi(a \mid y_j^{\text{tot}}) = \pi(a \mid y) + \alpha \sum_{j \in \mathcal{R}(y)} w_j\, \Delta_j^{\text{tot}}(a). \tag{1}$$

The score is the answer probability plus α times the average total effect of the answer-relevant hops on that answer. Redundant hops ($j \notin \mathcal{R}(y)$) are never contrasted.

Evidence-level rule (L1). While emitting the *result* tokens of a hop, we contrast the real context P with a type-matched counterfactual $\tilde{P}$ of identical length and entity types in which the relational content differs (on CLUTRR: names permuted within each gender group, so every sentence and entity type is kept but who-is-related-to-whom changes; we evaluate L1 on CLUTRR only, where $\tilde{P}$ is exactly type-matched): $t^\star = \arg\max_{t \in \mathcal{V}_\beta} (1+\alpha)\pi(t \mid P, \cdot) - \alpha\pi(t \mid \tilde{P}, \cdot)$. Unlike an empty context, $\tilde{P}$ is in-distribution and controls for length and type cues.

Remark 1 (What each level can fix)**.** *L2 makes the answer depend* more *on the chain; it therefore repairs* readout *errors (chain correct, answer wrong because a shortcut or distractor dominates the readout) but cannot repair a wrong chain—it may even commit harder to one. Chain errors are the target of L1, which acts while the hops are produced.*

Cost. L2 needs $1 + |\mathcal{R}(y)|$ prefix forwards and a KV-cached pass over the candidate tokens; no extra generation. L1 doubles the forward cost of result tokens only.

Choice of α. α is selected on the CLUTRR validation split and frozen (ties broken toward the smaller α, so HGCD reduces to CoT absent evidence); an adaptive variant sets $\alpha \cdot \mathrm{JSD}(p, \bar{p}_j)$ as in AdaCAD [27].

Propositions for the decoding probe

These propositions rest on a mixture-reader assumption that our data do not support (R5); we keep them because they make the probe's intended mechanism explicit.

Assumption 2 (Mixture reader)**.** $\pi(a \mid Q, P, y) = \lambda_c \pi_c(a \mid y, P_{\text{sup}}) + \lambda_s \pi_s(a \mid Q) + \lambda_d \pi_d(a \mid P_{\text{dist}})$*, with* λ*'s non-negative, summing to one, and invariant to interventions on hops.*

Proposition 3 (Exact cancellation in probability space)**.** *Under assumption 2,*

$$s_\alpha(a) = \pi(a \mid y) + \alpha\lambda_c \sum_j w_j \big[\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})\big],$$

i.e. the shortcut π_s *and distractor* π_d *components cancel in the contrast for every* α*. The log-space analogue does not cancel them.*

Proposition 4 (CAD amplifies distractor paths)**.** *Under assumption 2 applied to direct answering, with no-context distribution* $\pi_s(a \mid Q)$*, CAD's score is* $(1{+}\alpha)\log\big[\lambda_c\pi_c{+}\lambda_s\pi_s{+}\lambda_d\pi_d\big](a){-}\alpha\log\pi_s(a \mid Q)$*. For any* $\alpha > 0$ *and any gold answer* g*, a distractor answer* a_d *with* $\pi_d(a_d) = 1$ *is preferred by CAD whenever* $\pi_s(a_d \mid Q)$ *is small enough, even if* $\pi(g \mid Q, P) > \pi(a_d \mid Q, P)$*. In contrast, HGCD's contrast term does not depend on* π_d *at all (proposition 3).*

Prediction. Under distractor injection, CAD's accuracy degrades more than HGCD's, i.e. $|\text{ATE}_{\text{distract}}|$ is larger for CAD.

Proposition 5 (PNS limit)**.** *Interpret* $\pi(a \mid y_j^{\text{tot}})$ *as* $P(A{=}a \mid \text{do}(H_j{=}\perp))$ *and assume exogeneity and monotonicity as in Yu et al. [30]. Then* $\Delta_j^{\text{tot}}(a) = \text{PNS}_j(a)$ *and* $\lim_{\alpha\to\infty} s_\alpha(a)/\alpha = \sum_{j\in\mathcal{R}(y)} w_j \,\text{PNS}_j(a)$*. Hence, within* $\mathcal{V}_\beta$*, HGCD interpolates between the most probable answer (*$\alpha{=}0$*) and the answer for which the relevant hops are, on average, most necessary and sufficient.*

B Proofs

Proposition 1. y_j^{nai} changes only h_j and keeps $h_{j+1:k}$ at their observed values, so Δ_j^{nai} is by definition the controlled direct effect $\pi(a \mid h_j, h_{-j}) - \pi(a \mid \perp, h_{-j})$ [13]. Under assumption 1, $\pi(a \mid h_{1:k}) = g(a \mid h_{j+1:k})$ for $j < k$, so both terms equal $g(a \mid h_{j+1:k})$ and the difference is zero. y_j^{tot} additionally changes $h_{j+1:k}$ (their results are masked), so g is evaluated at a different argument and the difference is unconstrained. On compositional chains π reads the relation of every hop, so removing h_j alone changes the composed relation. □

Proposition 3. Substituting assumption 2 into eq. (1), with $\sum_j w_j = 1$ and π_s, π_d unaffected by interventions on hops: $s_\alpha(a) = (1{+}\alpha)[\lambda_c\pi_c(a \mid y) + \lambda_s\pi_s + \lambda_d\pi_d] - \alpha\sum_j w_j[\lambda_c\pi_c(a \mid y_j^{\text{tot}}) + \lambda_s\pi_s + \lambda_d\pi_d] = \pi(a \mid y) + \alpha\lambda_c\sum_j w_j[\pi_c(a \mid y) - \pi_c(a \mid y_j^{\text{tot}})]$. In log space, $\log(\lambda_c u + r) - \log(\lambda_c v + r)$ depends on $r = \lambda_s\pi_s + \lambda_d\pi_d$ unless $u = v$, so the shortcut terms do not cancel. □

Proposition 4. Let $m(a) = \pi(a \mid Q, P)$. CAD prefers a_d over g iff $(1{+}\alpha)\log\frac{m(a_d)}{m(g)} > \alpha\log\frac{\pi_s(a_d)}{\pi_s(g)}$. The left side is bounded below by $(1{+}\alpha)\log\frac{\lambda_d}{m(g)}$ since $m(a_d) \geq \lambda_d$, while the right side tends to $-\infty$ as $\pi_s(a_d) \to 0$. For HGCD, both terms of eq. (1) condition on the same P, so $\lambda_d\pi_d$ enters both and cancels (proposition 3). □

Proposition 5. Under exogeneity and monotonicity, Lemma 2 of Yu et al. [30] gives PNS = $P(A{=}a \mid \mathrm{do}(S)) - P(A{=}a \mid \mathrm{do}(S'))$. With $S = y$ and $S' = y_j^{\mathrm{tot}}$ this is $\Delta_j^{\mathrm{tot}}(a)$. Dividing eq. (1) by α and letting $\alpha \to \infty$ removes the $\pi(a \mid y)/\alpha$ term. □

C Implementation and Reproducibility

Candidate log-probabilities are computed with one prefix forward and a KV-cached pass over the candidate tokens (verified against full recomputation to 2×10^{-5}). Prompts are two-shot, hop-structured; the second CLUTRR shot uses a relation outside the label set to avoid leaking label statistics. All interventions are seeded functions of the example id. The Kaggle notebook embeds the full package and writes every table in this paper.

D SmolLM2-1.7B tables

Estimator	clutrr		clutrr_copy		musique	
	all	PS=1	all	PS=1	all	PS=1
Mask hop only (direct)	0.540	0.516	0.643	0.645	0.682	0.696
Mask hop + DRG descendants (total)	0.559	0.559	0.991	0.996	0.988	0.992
Replace result, keep downstream (direct)	0.503	0.474	0.651	0.643	0.658	0.658
Replace result, propagate (total)	0.532	0.494	0.996	1.000	0.987	0.993
DRG path indicator (structure only)	1.000	1.000	0.500	0.500	0.994	0.993
Placebo rewrite (control)	0.548	0.548	0.388	0.386	0.513	0.513
Mask token, content kept (control)	0.508	0.480	0.535	0.544	0.500	0.500

Intervention	Method	TE	NDE (chain fixed)	NIE (via chain)
dist8	Direct (no CoT)	-5.8 [-11.7,0.0]	-6.7 [-11.7,-1.7]	0.8 [-1.7,4.2]
dist8	CoT	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	CAD	-4.2 [-9.2,0.8]	-5.0 [-9.2,-1.7]	0.8 [-1.7,4.2]
dist8	AdaCAD	-3.3 [-8.3,0.8]	-5.0 [-10.0,-0.8]	1.7 [-1.7,5.0]
dist8	CAD-chain	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	Calibration (content-free)	-10.8 [-17.5,-3.3]	-1.7 [-5.0,1.7]	-9.2 [-15.8,-2.5]
dist8	Direct × CoT (PoE)	-7.5 [-14.2,0.0]	-0.8 [-4.2,1.7]	-6.7 [-13.3,0.0]
dist8	HGCD (no graph)	-8.3 [-15.0,-1.7]	0.8 [-2.5,4.2]	-9.2 [-15.8,-2.5]
dist8	HGCD (log-space)	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	HGCD (adaptive α)	-7.5 [-14.2,-0.8]	1.7 [-1.7,5.0]	-9.2 [-15.8,-2.5]
dist8	**HGCD**	-8.3 [-15.0,-1.7]	0.8 [-2.5,4.2]	-9.2 [-15.8,-2.5]
order	Direct (no CoT)	-2.5 [-7.5,2.5]	-0.8 [-4.2,2.5]	-1.7 [-5.8,2.5]
order	CoT	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	CAD	-0.8 [-5.0,3.3]	0.0 [-4.2,4.2]	-0.8 [-4.2,2.5]
order	AdaCAD	-0.8 [-4.2,2.5]	-0.8 [-4.2,2.5]	0.0 [-3.3,3.3]
order	CAD-chain	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	Calibration (content-free)	0.0 [-7.5,7.5]	-0.8 [-2.5,0.0]	0.8 [-6.7,8.3]
order	Direct × CoT (PoE)	1.7 [-5.8,9.2]	-0.8 [-4.2,1.7]	2.5 [-4.2,10.0]
order	HGCD (no graph)	3.3 [-5.0,11.7]	0.8 [-1.7,3.3]	2.5 [-5.8,10.8]
order	HGCD (log-space)	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	HGCD (adaptive α)	2.5 [-5.0,10.0]	0.8 [-1.7,3.3]	1.7 [-5.8,10.0]
order	**HGCD**	3.3 [-5.0,11.7]	0.8 [-1.7,3.3]	2.5 [-5.8,10.8]